TONGJI CALCULUS, 7TH EDITION · CHAPTER 2

DERIVATIVES & DIFFERENTIALS EXERCISE WORKBOOK

100 original and textbook-method adaptations · basic to challenge

Goodnotes-optimized 4:3 edition · 2026

How to use this set

100	All 5 sections	Basic → challenge	8 mixed formats
Questions	Sections	Progression	Formats

Work through Foundation, Methods, Synthesis, and Challenge in that order. Each question shows its difficulty, format, and suggested time. Attempt it before opening the solution book; record the cause of each error and retry after 48 hours.

“Textbook-method adaptation” means the representative method is retained while the structure, parameters, or task have been independently redesigned.

Scope: Chapter 2 tools only. Mean value theorems, L'Hopital's rule, Taylor expansions, monotonicity or extremum tests, curvature, integration, and power series are not used.

Coverage matrix

Sec.	Topic	Items	Key coverage
1	The Derivative Concept	24	definition, one-sided derivatives, geometry, rates, continuity
2	Differentiation Rules	28	algebraic, chain, inverse, basic formulas, parameters
3	Higher Derivatives	16	second and nth derivatives, cycles, induction, products
4	Implicit and Parametric Derivatives; Related Rates	20	implicit, parametric, related rates, units
5	Differentials	12	definition, rules, linear approximation, error estimates

Foundation

Attempt independently before consulting the solutions.

§1 · The Derivative Concept	Basic	Single choice	4 min	Textbook-method adaptation

Q001 · Recognizing the Definition of a Derivative

Suppose f is defined in a neighborhood of x_0. Which limit, if it exists, equals $f'(x_0)$?

A. A. lim_(Δx→0)[f(x_0+Δx)-f(x_0)]/Δx

B. B. lim_(Δx→0)[f(x_0+Δx)-f(x_0)]/x_0

C. C. lim_(x→x_0)[f(x)-f(x_0)]/x

D. D. lim_(Δx→0)[f(x_0+Δx)+f(x_0)]/Δx

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Basic	Fill in the blank	4 min	Textbook-method adaptation

Q002 · Expressing Instantaneous Velocity as a Limit

A particle moves along a line with position $s=s(t)$. Its instantaneous velocity at $t=3$ can be written as $v(3)=$______.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Basic	Calculation	7 min	Textbook-method adaptation

Q003 · Differentiate at a Nonzero Point from First Principles

Using only the definition of the derivative, find the derivative of $f(x)=x^2$ at $x=2$.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Basic	True/false with justification	5 min	Textbook-method adaptation

Q004 · Does Differentiability Guarantee Continuity?

Decide whether the statement is true and justify: if f is differentiable at x_0, then f is continuous at x_0.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Basic	Single choice	4 min	Textbook-method adaptation

Q005 · Writing a Tangent Line from a Derivative

Suppose $f(2)=5$ and $f'(2)=-3$. Which equation is the tangent line to $y=f(x)$ at $(2,5)$?

A. A. $y-5=-3(x-2)$

B. B. $y-2=-3(x-5)$

C. C. $y-5=(1/3)(x-2)$

D. D. $y+5=-3(x+2)$

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Basic	Fill in the blank	6 min	Textbook-method adaptation

Q006 · One-Sided Derivatives of an Absolute-Value Function

Let $f(x)=|x|$. Then $f'_{-}(0)=$______, $f'_{+}(0)=$______, so f is ______ at 0 (write “differentiable” or “not differentiable”).

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Basic	Calculation	8 min	Textbook-method adaptation

Q007 · Tangent and Normal Lines from First Principles

Let $P(1,1)$ lie on the curve $y=x^2$. Use the definition of the derivative to find the tangent and normal lines at P.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Basic	Single choice	4 min	Textbook-method adaptation

Q008 · Choose the Derivative of a Logarithm

For $x>0$, what is the derivative of $y=\ln x$?

A. A. $y'=1/x$

B. B. $y'=\ln x$

C. C. $y'=x$

D. D. $y'=e^x$

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Basic	Single choice	4 min	Textbook-method adaptation

Q009 · A Polynomial Times an Exponential

Let $f(x)=x^2e^x$. Which expression equals $f'(x)$?

A. A. $2xe^x$

B. B. $e^x(x^2+2x)$

C. C. x^2e^x

D. D. $e^x(x+2)$

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Basic	True/false with justification	5 min

Q010 · Can a Quotient Be Differentiated Termwise?

True or false, with justification: If u and v are differentiable and $v'(x) \neq 0$, then $[u(x)/v(x)]' = u'(x)/v'(x)$.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Basic	Fill in the blank	5 min	Textbook-method adaptation

Q011 · A Linear Combination of Three Basic Functions

For $x>0$, if $y=3e^x-2\ln x+5\arctan x$, then $y'=$________.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Basic	Calculation	6 min	Textbook-method adaptation

Q012 · The Fifth Power of a Linear Function

Differentiate $y=(2x-1)^5$ and explicitly identify the outer and inner functions.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Basic	Calculation	6 min	Textbook-method adaptation

Q013 · A Sine Divided by an Exponential

Differentiate $y=\sin x/e^{x}$ and write the result in the simplest product form.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Basic	Single choice	5 min	Textbook-method adaptation

Q014 · The Derivative of an Inverse at Corresponding Points

Suppose f has a differentiable inverse $g=f^{-1}$ near 2, with $f(2)=5$ and $f'(2)=-3$. What is $g'(5)$?

A. A. -3

B. B. $-1/3$

C. C. $1/3$

D. D. 3

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Basic	True/false with justification	4 min	Textbook-method adaptation

Q015 · Higher Derivatives of a Polynomial

True or false, with justification: If $y=x^4-3x+2$, then $y^{(5)}(x)\equiv 0$.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Basic	Fill in the blank	5 min	Textbook-method adaptation

Q016 · The 2026th Derivative of Cosine

If $y=\cos x$, then $y^{(2026)}=$________.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Basic	Single choice	5 min	Textbook-method adaptation

Q017 · Recognizing a First Implicit Derivative

On the curve $x^2+xy+y^2=7$, wherever y is a differentiable function of x, which expression equals y'?

A. A. $-(2x+y)/(x+2y)$

B. B. $-(2x+y)/(2x+y)$

C. C. $(2x+y)/(x+2y)$

D. D. $-(x+2y)/(2x+y)$

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Basic	Calculation	7 min	Textbook-method adaptation

Q018 · The Tangent to a Circle

At the point (3,4) on $x^2+y^2=25$, find dy/dx and the equation of the tangent line.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Basic	Fill in the blank	6 min	Textbook-method adaptation

Q019 · Evaluating an Implicit Derivative at a Point

For $x^2+xy=6$, dy/dx at $(2,1)$ is ____.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Basic	True/false with justification	8 min	Textbook-method adaptation

Q020 · The Domain of an Implicit-Derivative Formula

True or false, with justification: From $x^2+y^2=1$ we obtain $y'=-x/y$, so this formula gives a finite tangent slope at every point of the circle.

WORKSPACE · add pages as needed

§5 · Differentials	Basic	Single choice	5 min	Textbook-method adaptation

Q021 · Recognizing the Differential of a Function

If $y=f(x)$ is differentiable at x and $dx=\Delta x$, which expression is the differential dy?

A. A. $dy=f'(x)dx$

B. B. $dy=\Delta y$ for every finite Δx

C. C. $dy=f(x)dx$

D. D. dy is independent of dx

WORKSPACE · add pages as needed

§5 · Differentials	Basic	Fill in the blank	5 min	Textbook-method adaptation

Q022 · Numerical Differential of a Power Function

If $y=x^3$, then at $x=2$ with $dx=0.01$, $dy=$____.

WORKSPACE · add pages as needed

Methods

Attempt independently before consulting the solutions.

§1 · The Derivative Concept	Standard	Calculation	8 min	Textbook-method adaptation

Q023 · Derivative of the Reciprocal Function at Any Nonzero Point

Let $f(x)=1/x$. For any $a\neq 0$, use only the definition of the derivative to find $f'(a)$.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Standard	Multiple choice	7 min

Q024 · Logical Relations Involving Differentiability

Let x_0 be an interior point of the domain of f. Which statements are correct?

A. A. If $f'(x_0)$ exists, then $f'_-(x_0)$ and $f'_+(x_0)$ both exist and are equal

B. B. If f is continuous at x_0, then $f'(x_0)$ must exist

C. C. If $f'(x_0)$ exists, then f is continuous at x_0

D. D. If $f'_-(x_0)$ and $f'_+(x_0)$ are both finite, then $f'(x_0)$ must exist

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Standard	Proof	10 min	Textbook-method adaptation

Q025 · Prove Differentiability Implies Continuity via an Infinitesimal Remainder

Suppose f is differentiable at x_0. Prove that some $\alpha(\Delta x)\to 0$ satisfies $f(x_0+\Delta x)-f(x_0)=f'(x_0)\Delta x+\alpha(\Delta x)\Delta x$, and use it to prove continuity at x_0.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Standard	Calculation	10 min	Textbook-method adaptation

Q026 · Matching Parameters at a Junction

Let $f(x)=\{x^2,\ x\le 1;\ ax+b,\ x>1\}$. Find a,b so that f is differentiable at $x=1$, and find $f'(1)$.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Standard	Fill in the blank	8 min	Textbook-method adaptation

Q027 · Instantaneous Velocity at a Specified Time

A particle has position $s(t)=t^3-6t^2+9t$ meters, with t in seconds. Use a difference-quotient limit to find the instantaneous velocity at $t=2$, namely $v(2)=$______.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Standard	Calculation	10 min	Textbook-method adaptation

Q028 · Tangent and Normal to a Square-Root Curve

Without using a ready-made derivative formula, find the derivative of $y=\sqrt{x}$ at $(4,2)$ from the definition, and write the tangent and normal lines.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Standard	True/false with justification	6 min	Textbook-method adaptation

Q029 · Testing the Converse of Differentiability Implies Continuity

Decide whether the statement is true and justify rigorously: if f is continuous at x_0, then f is differentiable at x_0.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Standard	Proof	8 min	Textbook-method adaptation

Q030 · Prove a Translated Absolute Value Is Not Differentiable at Its Corner

Let a be any real number. Prove that $f(x)=|x-a|$ is continuous at $x=a$, but has left derivative -1 and right derivative 1 there, and hence is not differentiable.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Calculation	7 min	Textbook-method adaptation

Q031 · Product Rule and Simplification

Differentiate $y=(x^2+1)(x^3-2x)$, and simplify the result to a polynomial.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Fill in the blank	6 min	Textbook-method adaptation

Q032 · Identify the Inner and Outer Functions

If $y=\sin(3x^2-1)$, then $y'=$______.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Multiple choice	8 min

Q033 · Distinguish Four Differentiation Formulas

Assume all differentiability, domain, nonzero-denominator, and inverse-continuity conditions required by the formulas hold. For the inverse formula, let $y_0=f(x_0)$ and $f'(x_0)\neq 0$. Which formulas are correct?

A. A. $(fg)'=f'g+fg'$

B. B. $(f/g)'=(f'g-fg')/g^2$ $(g\neq 0)$

C. C. $(f\circ g)'(x)=f'(x)g'(x)$

D. D. $(f^{-1})'(y_0)=1/f'(x_0)$

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Calculation	7 min	Textbook-method adaptation

Q034 · Quotient Rule for a Rational Function

Differentiate $y=(x^2+1)/(x-1)$, and state where the derivative formula applies.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Calculation	9 min	Textbook-method adaptation

Q035 · A Product Nested Inside a Quotient

For $x>0$ and $x\neq 1$, differentiate $y=(x^2+1)\ln x/(x-1)$, and give a result over a common denominator.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Single choice	6 min	Textbook-method adaptation

Q036 · A Composite of Arcsine and a Square Root

For $0<x<1$, which expression is the derivative of $y=\arcsin\sqrt{x}$?

- A. A. $1/\sqrt{(1-x)}$
- B. B. $1/[2\sqrt{x}\sqrt{(1-x)}]$
- C. C. $1/[2\sqrt{x}(1-x)]$
- D. D. $-1/[2\sqrt{x}\sqrt{(1-x)}]$

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Multiple choice	7 min

Q037 · Distinguishing Four Differentiation Rules

Assume the functions involved are differentiable at the relevant points and all expressions are defined. Which statements are correct?

A. A. $[u(v(x))]'=u'(v(x))v'(x)$

B. B. $[u(x)v(x)]'=u'(x)v'(x)$

C. C. $[u(x)/v(x)]'=[u'(x)v(x)-u(x)v'(x)]/v^2(x)$, where $v(x)\neq 0$

D. D. If f has a differentiable inverse and $f'(f^{-1}(y))\neq 0$, then $(f^{-1})'(y)=1/f'(f^{-1}(y))$

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Calculation	8 min	Textbook-method adaptation

Q038 · A Trigonometric Square Inside a Logarithm

Differentiate $y=\ln[1+\sin^2(3x)]$, listing the layers from outside to inside.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Calculation	9 min	Textbook-method adaptation

Q039 · Both Base and Exponent Depend on x

Differentiate $y=(x^2+1)^{\sin x}$ using logarithmic differentiation, and state its domain.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Proof	13 min	Textbook-method adaptation

Q040 · Proving the Inverse Derivative Formula from the Definition

Suppose f is differentiable at x_0 with $f'(x_0)\neq 0$ and is one-to-one near x_0. Its inverse g is continuous at $y_0=f(x_0)$. Prove from the derivative definition that $g'(y_0)=1/f'(x_0)$.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Standard	Fill in the blank	8 min	Textbook-method adaptation

Q041 · Determining Two Parameters in a Piecewise Function

Let $f(x)=\{ax+b \text{ for } x<1;\ x^2+1 \text{ for } x\geq 1\}$. For f to be differentiable at $x=1$, a=________ and b=________.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Standard	Calculation	9 min	Textbook-method adaptation

Q042 · Second and Third Derivatives of an Exponential-Cosine Product

Let $y=e^{2x}\cos x$. Find y'' and y'''.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Standard	True/false with justification	7 min

Q043 · Does the nth Derivative of a Product Have Only Two Terms?

True or false, with justification: For every $n \geq 2$, if u and v have the needed derivatives, then $(uv)^{(n)} = u^{(n)}v + uv^{(n)}$.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Standard	Calculation	7 min	Textbook-method adaptation

Q044 · Higher Derivatives of an Exponential Plus a Sine

Let $y=\sin(2x)+e^{3x}$. Find y'' and y'''.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Standard	Proof	13 min	Textbook-method adaptation

Q045 · The nth Derivative of the Reciprocal of a Linear Factor

For every integer $n \geq 0$, prove by induction that if $y=1/(x-a)$, $x \neq a$, then $y^{(n)}=(-1)^n n!/(x-a)^{n+1}$.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Standard	Calculation	11 min	Textbook-method adaptation

Q046 · First Derivative of an Exponential Implicit Curve

The curve e^(xy)+x+y=1 passes through (0,0). Find the general expression for y′ and the tangent slope at that point.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Standard	Calculation	11 min

Q047 · Differentiating a Trigonometric Implicit Curve

The curve $\sin(x+y)=xy$ passes through $(0,0)$. Find the general expression for y' and the tangent line at that point.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Standard	Calculation	9 min	Textbook-method adaptation

Q048 · First Derivative and Tangent of a Parametric Curve

For $x=t^2+1$ and $y=t^3-t$, find dy/dx at $t=1$ and write the tangent line.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Standard	Fill in the blank	6 min	Textbook-method adaptation

Q049 · The Parametric Derivative Formula and Its Value

If $x=t^2$ and $y=t^4+t$, then when $dx/dt\neq 0$, $dy/dx=$____; at $t=1$ its value is ____.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Standard	Single choice	6 min

Q050 · Condition for a Vertical Tangent of a Parametric Curve

Suppose the parameter derivatives exist at $t=t_0$. Which condition most directly indicates a vertical tangent?

A. A. $dx/dt=0$ and $dy/dt\neq0$

B. B. $dx/dt\neq0$ and $dy/dt=0$

C. C. $dx/dt=dy/dt=1$

D. D. $dx/dt=0$ and $dy/dt=0$, with no further analysis needed

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Standard	Calculation	9 min	Textbook-method adaptation

Q051 · Volume Rate of an Expanding Sphere

The radius of a spherical balloon increases at 2 cm/s. How fast is its volume increasing when the radius is 5 cm?

WORKSPACE · add pages as needed

§5 · Differentials	Standard	True/false with justification	9 min	Textbook-method adaptation

Q052 · Differentiability and Existence of a Derivative in One Variable

True or false, with justification: For a one-variable function $y=f(x)$, differentiability at a point is equivalent to existence of the derivative there, and the differential coefficient is $f'(x)$.

WORKSPACE · add pages as needed

§5 · Differentials	Standard	Calculation	8 min	Textbook-method adaptation

Q053 · Differential of a Logarithmic Composite

Let $y=\ln(1+x^2)$. Find dy, and evaluate it at $x=1$ with $dx=0.02$.

WORKSPACE · add pages as needed

§5 · Differentials	Standard	Calculation	8 min	Textbook-method adaptation

Q054 · Approximating a Square Root with Differentials

Using only differential linearization near $x_0=4$, approximate $\sqrt{4.04}$.

WORKSPACE · add pages as needed

§5 · Differentials	Standard	Multiple choice	9 min

Q055 · Correct Statements about Differential Estimates

Let $y=f(x)$ be differentiable at x with $y\neq0$. Which statements are correct? Select all that apply.

A. A. As $\Delta x\to0$, $\Delta y=dy+o(\Delta x)$

B. B. dy is linear in dx

C. C. $\Delta y=dy$ for every finite dx

D. D. dy/y is a first-order estimate of the relative increment $\Delta y/y$

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Advanced	Calculation	12 min

Q056 · Differentiability of an Oscillatory Function at the Origin

Let $f(x)=\{x^2\sin(1/x),\ x\neq 0;\ 0,\ x=0\}$. Decide whether f is differentiable at 0, and find the complete derivative $f'(x)$.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Advanced	Proof	10 min	Textbook-method adaptation

Q057 · Slope of an Even Function at the Origin

Suppose f is even and differentiable at $x=0$. Prove that $f'(0)=0$.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Advanced	Single choice	8 min

Q058 · Recognizing the Limit of an Asymmetric Secant

Suppose f is differentiable at x_0. Which limit must equal $f'(x_0)$?

A. A. $\lim_{h\to 0}[f(x_0+2h)-f(x_0-h)]/(3h)$

B. B. $\lim_{h\to 0}[f(x_0+2h)-f(x_0-h)]/(2h)$

C. C. $\lim_{h\to 0}[f(x_0+h)-f(x_0-h)]/h$

D. D. $\lim_{h\to 0}[f(x_0+2h)-f(x_0)]/h$

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Advanced	Calculation	12 min

Q059 · A Moving Corner and Parameter Classification

Let $f_{(a,b)}(x)=|x-a|+bx$, where $a,b\in\mathbb{R}$. Determine all pairs (a,b) for which $f_{(a,b)}$ is differentiable at $x=0$, and find $f'_{(a,b)}(0)$.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Advanced	Calculation	8 min	Textbook-method adaptation

Q060 · Derivative of an Inverse at a Corresponding Point

Let $f(x)=x^3+x$ and $g=f^{-1}$. Given that g is differentiable at $y=2$, find $g'(2)$.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Advanced	Proof	14 min	Textbook-method adaptation

Q061 · Prove the Product Rule from First Principles

Suppose u and v are differentiable at x_0. Using only the definition of the derivative, prove $(uv)'(x_0)=u'(x_0)v(x_0)+u(x_0)v'(x_0)$.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Advanced	Calculation	12 min	Textbook-method adaptation

Q062 · Combined Multilayer Chain and Quotient Rules

Differentiate $y=e^{\sin(x^2)}/(1+x^3)$, and state its domain.

WORKSPACE · add pages as needed

Synthesis

Attempt independently before consulting the solutions.

§1 · The Derivative Concept	Advanced	Parameter / synthesis / application	14 min	Textbook-method adaptation

Q063 · From Average Velocity to Instantaneous Direction

A particle has position $s(t)=t^3-3t^2+2t$ meters, with t in seconds. (1) Find the average velocity on $[1,1+h]$ for $h\neq 0$. (2) Use the definition to find $v(1)$. (3) Write the tangent to the position-time curve at $(1,s(1))$. (4) Interpret the sign of $v(1)$.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Advanced	Error diagnosis	10 min

Q064 · The Error in an "Infinite Derivative"

The function $f(x)=\sqrt{x}$ has domain $[0,+\infty)$. A student writes: "$f'(0)=\lim_{h\to 0}[\sqrt{h}-0]/h=\lim_{h\to 0}1/\sqrt{h}=+\infty$, so $f'(0)=+\infty$." Identify every key error and give the correct conclusion.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Advanced	Multiple choice	8 min

Q065 · Checking Both Formula and Domain

Which differentiation formulas are correct on the indicated domains?

A. A. $(\ln|x|)' = 1/x,\ x \neq 0$

B. B. $(\arccos x)' = 1/\sqrt{(1-x^2)},\ -1<x<1$

C. C. $(a^x)' = a^x \ln a,\ a>0$ and $a \neq 1$

D. D. $[\sqrt{(1-x^2)}]' = -x/\sqrt{(1-x^2)},\ -1<x<1$

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Advanced	Calculation	11 min	Textbook-method adaptation

Q066 · Chain Rule for a Four-Layer Composite

Differentiate $y=\arctan[e^{\sin(x^2)}]$, using layer variables to explain the source of every factor.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Advanced	Proof	12 min	Textbook-method adaptation

Q067 · Proving the Derivative Rule for Real Powers

Let α be any real number. Using $x^\alpha=e^{\alpha\ln x}$, prove that $(x^\alpha)'=\alpha x^{\alpha-1}$ for $x>0$.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Advanced	Parameter / synthesis / application	15 min	Textbook-method adaptation

Q068 · Differentiability Classification for a Three-Parameter Piecewise Function

Let $f(x)=\{x^2+ax+b \text{ for } x<1;\ c \ln x+2 \text{ for } x\geq 1\}$. Find all real triples (a,b,c) for which f is differentiable at $x=1$, and give $f'(1)$.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Advanced	Proof	12 min	Textbook-method adaptation

Q069 · Deriving the Derivative of Arctangent from the Inverse Formula

View arctan x as the inverse of tan y on $(-\pi/2,\pi/2)$, and prove that $(\arctan x)'=1/(1+x^2)$.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Advanced	Calculation	10 min	Textbook-method adaptation

Q070 · The nth Derivative of the Natural Logarithm

For $x>0$, compute the first four derivatives of $\ln x$ and then state its nth derivative for $n\geq 1$.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Advanced	Proof	18 min	Textbook-method adaptation

Q071 · A Unified nth-Derivative Formula for an Exponential-Cosine Product

Let $a^2+b^2>0$, $r=\sqrt{(a^2+b^2)}$, and choose θ so that $\cos\theta=a/r$ and $\sin\theta=b/r$. Prove by induction that $y=e^{ax}\cos(bx+\varphi)$ satisfies $y^{(n)}=r^n e^{ax}\cos(bx+\varphi+n\theta)$.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Advanced	Parameter / synthesis / application	16 min	Textbook-method adaptation

Q072 · The General Derivative of $x^2\sin x$

Let $y=x^2\sin x$. For $n\geq 2$, find an explicit formula for $y^{(n)}$ and check it at $n=2$.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Advanced	Calculation	13 min	Textbook-method adaptation

Q073 · Decompose Before Taking Higher Derivatives of a Rational Function

Let $y=1/[(x-1)(x+2)]$, $x\neq 1,-2$. Find $y^{(n)}$ for $n\geq 0$.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Advanced	Calculation	18 min	Textbook-method adaptation

Q074 · Second Derivative of a Quadratic Implicit Curve

The curve $x^2+xy+y^2=7$ passes through $(1,2)$. Find y' and y'' at that point.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Advanced	Calculation	14 min	Textbook-method adaptation

Q075 · Second Derivative of a Parametric Curve

For $x=t^2+1$ and $y=t^3$, find dy/dx and d^2y/dx^2 at $t=1$.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Advanced	Multiple choice	9 min

Q076 · Modeling Principles for Related Rates

Which statements about related-rate problems are correct? Select all that apply.

A. A. First write a geometric or physical relation valid over time, then differentiate with respect to t

B. B. If a length is decreasing, its rate is negative under the chosen positive direction

C. C. The unit of dV/dt is length³/time

D. D. Current numerical values may be substituted before differentiation and treated as constants because only one instant matters

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Advanced	Parameter / synthesis / application	18 min	Textbook-method adaptation

Q077 · Radius and Volume Rates in a Conical Water Surface

Water in an inverted conical tank forms similar cones satisfying $h=3r$, with h and r in cm. The depth increases at $dh/dt=6$ cm/min. When $h=6$ cm, find dr/dt and dV/dt.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Advanced	Parameter / synthesis / application	20 min	Textbook-method adaptation

Q078 · Linear and Angular Rates of a Sliding Ladder

A 10 m ladder leans against a wall. Its foot moves away from the wall at $dx/dt=0.6$ m/s. Let x be the foot's distance from the wall, y the top height, and θ the angle with the ground. When $x=6$ m, find dy/dt and $d\theta/dt$.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Advanced	Proof	17 min	Textbook-method adaptation

Q079 · A General Derivative Formula for a Separable Implicit Equation

Let φ and ψ be differentiable, and suppose $y=y(x)$ satisfies $\varphi(x)+\psi(y)=C$ with $\psi'(y)\neq 0$. Prove $y'=-\varphi'(x)/\psi'(y)$, and explain the role of $\psi'(y)\neq 0$.

WORKSPACE · add pages as needed

§5 · Differentials	Advanced	Calculation	13 min	Textbook-method adaptation

Q080 · Absolute and Relative Error Estimates for a Circle's Area

A circle's radius is measured as $r=10$ cm with possible absolute measurement error $|dr|\le 0.02$ cm. Use differentials to estimate the maximum absolute and relative errors in $A=\pi r^2$.

WORKSPACE · add pages as needed

§5 · Differentials	Advanced	Calculation	11 min	Textbook-method adaptation

Q081 · Approximating a Reciprocal Square Root

Use differential linearization at $x_0=1$ to approximate $1/\sqrt{0.98}$, and state whether this method provides a rigorous error bound.

WORKSPACE · add pages as needed

§5 · Differentials	Advanced	Parameter / synthesis / application	18 min	Textbook-method adaptation

Q082 · Linear Prediction and Actual Discrepancy for a Fifth Power

Let $y=x^5$. Using $x_0=1$ and $dx=0.02$, approximate $(1.02)^5$ by differentials and give the predicted relative increment. Then use the exact value 1.1040808032 from direct multiplication to find exact minus approximate, and explain why this is not an error bound supplied by this chapter.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Hard	True/false with justification	12 min	Textbook-method adaptation

Q083 · Complete Classification of a Parameterized Power Corner

Decide whether the statement is true and prove it: for $\alpha>0$, $f_\alpha(x)=|x|^\alpha$ is differentiable at $x=0$ if and only if $\alpha>1$; when differentiable, $f'_\alpha(0)=0$.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Hard	Parameter / synthesis / application	18 min

Q084 · Stable Secants but Uncontrolled Nearby Tangents

Let $f(x)=\{x^2\sin(1/x^2),\ x\neq0;\ 0,\ x=0\}$. (1) Prove continuity at 0. (2) Prove differentiability at 0 and write the tangent there. (3) Find $f'(x)$ for $x\neq0$. (4) Construct $x_n\to0^+$ showing that f' is not continuous at 0.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Hard	Calculation	14 min	Textbook-method adaptation

Q085 · Structural Simplification after Nested Rules

Differentiate $y=\arctan[(e^x-1)/(e^x+1)]$, and simplify the result so that it contains no compound fraction.

WORKSPACE · add pages as needed

§2 · Differentiation Rules	Hard	Proof	16 min	Textbook-method adaptation

Q086 · Prove the Inverse-Function Derivative Formula from First Principles

Suppose f is one-to-one on a neighborhood U of x_0, its image contains a neighborhood of $y_0=f(x_0)$, and it has inverse $x=g(y)$ on that image. Assume f is differentiable at x_0, $f'(x_0)\neq 0$, and $g(y)\to x_0$ as $y\to y_0$. Prove $g'(y_0)=1/f'(x_0)$.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Hard	Calculation	16 min	Textbook-method adaptation

Q087 · The nth Derivative of a Cubic Times an Exponential

Let $y=x^3e^{2x}$. For $n\geq 3$, find $y^{(n)}$ with no summation sign in the answer.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Hard	Proof	20 min	Textbook-method adaptation

Q088 · Inductive Proof of the Higher Product Formula

Suppose u and v have the required derivatives. Prove by induction that for $n \geq 0$, $(uv)^{(n)}=\Sigma\_(k=0)^{n}C(n,k)u^{(k)}v^{(n-k)}$.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Hard	Error diagnosis	14 min

Q089 · Diagnosing a Missing Repeated Chain Factor

For $y=\ln(2x-1)$, $x>1/2$, a student writes $y^{(n)}=(-1)^{n-1}(n-1)!/(2x-1)^n$. Identify the error, give the correct formula, and check at least $n=1$ and $n=2$.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Hard	Calculation	24 min	Textbook-method adaptation

Q090 · Second Derivative of a Cubic Implicit Curve

The curve $x^3+y^3=6xy$ passes through $(3,3)$. Find y' and y'' at that point.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Hard	Proof	23 min	Textbook-method adaptation

Q091 · Deriving the Second Parametric Derivative Formula

Let $x=x(t)$ and $y=y(t)$ be twice differentiable with $x'(t)\neq 0$. Prove $d^2y/dx^2=[x'(t)y''(t)-y'(t)x''(t)]/[x'(t)]^3$.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Hard	Error diagnosis	22 min	Textbook-method adaptation

Q092 · Diagnosing a Frozen Water-Surface Radius

An inverted conical tank is 9 m high with rim radius 3 m. The water depth rises at 0.2 m/min. Find the volume rate when $h=3$ m. A student fixes the water radius at 3 m, writes $V=(1/3)\pi\cdot 3^2 h$, and obtains $dV/dt=0.6\pi$ m^3/min. Diagnose the error and give the correct result.

WORKSPACE · add pages as needed

§5 · Differentials	Hard	Proof	24 min	Textbook-method adaptation

Q093 · Equivalence between a Linear Increment Decomposition and a Derivative

Prove that a one-variable function f is differentiable at x_0 if and only if there is a constant A such that $\Delta y=A\Delta x+o(\Delta x)$. Prove that $A=f'(x_0)$, so $dy=A\,dx$.

WORKSPACE · add pages as needed

§5 · Differentials	Hard	Error diagnosis	20 min	Textbook-method adaptation

Q094 · Diagnosing a Wrong Base Point and an Exact Equality

To approximate $\sqrt{25.5}$, a student takes $f(x)=\sqrt{x}$, $x_0=25$, and $\Delta x=0.5$, but writes $dy=f'(25.5)\Delta x=0.5/[2\sqrt{25.5}]\approx 0.04951$ and claims $\sqrt{25.5}=5.04951$. Identify all key errors, give the correct differential approximation, and state whether a rigorous error bound follows.

WORKSPACE · add pages as needed

Challenge

Attempt independently before consulting the solutions.

§3 · Higher Derivatives	Hard	Parameter / synthesis / application	17 min	Textbook-method adaptation

Q095 · Evaluating the 2026th Derivative at Zero

Let $y=e^x(\cos x+\sin x)$. Without expanding derivative by derivative, find $y^{(2026)}(0)$ and explain the cyclic structure used.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Hard	Calculation	17 min	Textbook-method adaptation

Q096 · The nth Derivative of a Rational Function with a Repeated Pole

Let $y=1/[x^2(x-1)]$, $x\neq 0,1$. Find $y^{(n)}$ for $n\geq 0$.

WORKSPACE · add pages as needed

§1 · The Derivative Concept	Challenge	Proof	28 min

Q097 · Critical Exponents in a Two-Parameter Oscillatory Family

Let $\alpha,\beta>0$ and $f_{(\alpha,\beta)}(x)=\{|x|^\alpha \sin(1/|x|^\beta),\ x\neq 0;\ 0,\ x=0\}$. (1) Prove that $f_{(\alpha,\beta)}$ is continuous at 0. (2) Completely classify differentiability at 0. (3) When it is differentiable, completely classify continuity of $f'_{(\alpha,\beta)}$ at 0.

WORKSPACE · add pages as needed

§3 · Higher Derivatives	Challenge	Proof	24 min

Q098 · The nth Derivative of a Squared Logarithm and Harmonic Numbers

Define $H_0=0$ and $H_m=1+1/2+\cdots+1/m$ for $m\geq 1$. For $x>0$, prove by induction that for $n\geq 1$, $[(\ln x)^2]^{(n)}=2(-1)^{n-1}(n-1)!x^{-n}[\ln x-H_{n-1}]$.

WORKSPACE · add pages as needed

§4 · Implicit and Parametric Derivatives; Related Rates	Challenge	Proof	32 min	Textbook-method adaptation

Q099 · A General Second-Derivative Formula for a Separable Implicit Equation

Let φ and ψ be twice differentiable, with $y=y(x)$ satisfying $\varphi(x)+\psi(y)=C$ and $\psi'(y)\neq0$. Prove $y''=-\{\varphi''(x)[\psi'(y)]^2+[\varphi'(x)]^2\psi''(y)\}/[\psi'(y)]^3$ using only the differentiation rules of this chapter.

WORKSPACE · add pages as needed

§5 · Differentials	Challenge	Proof	30 min	Textbook-method adaptation

Q100 · Proving the Product Rule for Differentials from the Definition

Let $u=u(x)$ and $v=v(x)$ be differentiable at x_0. Using only increment decompositions, prove that uv is differentiable at x_0 and $d(uv)=v\,du+u\,dv$. Do not cite the mean value theorem or later-chapter tools.

WORKSPACE · add pages as needed

Training value, limitations, and second-pass route

Strengths

- All five sections are covered and indexed by knowledge point.
- Definition, computation, proof, parameter, application, and error-diagnosis tasks expose more than formula substitution.
- Even the hard questions stay within Chapter 2 tools and emphasize first principles, rule selection, parameter classification, and rate models.
- Chinese and English identifiers and mathematics match one-to-one, supporting calculus vocabulary development.

Limitations

- One hundred questions cannot exhaust every composite-function pattern; weak categories may need extra drills.
- Textbook material is represented by independently rewritten method adaptations, not verbatim textbook questions.
- Difficulty is partly subjective and depends on algebra and trigonometric fluency.
- Differential approximations show first-order linearization without Taylor-style remainder bounds.
- To respect the Chapter 2 boundary, mean value theorems, L'Hopital, Taylor, and monotonicity or extremum tests are not used.

Recommended second pass

1. First pass: work under time limits, record confidence, and do not open the solutions.
2. Correction: classify each error as conceptual, algebraic, method-selection, or rigor/communication.
3. After 48 hours: redo wrong items and their neighbors without prompts.
4. After one week: sample by knowledge tag, emphasizing first principles, the chain rule, implicit conditions, rate units, and differential approximation.